Slides, posters, and figures

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Talks, posters, and figures carry the same research as a paper, but the reader sees them for less time and cannot reread them. This chapter gives practical rules for each.

Guidelines for slide decks

When preparing slides for a lab meeting, conference, qualifying exam, or thesis defense, follow these guidelines:

- **Always include slide numbers**: Slide numbers make feedback specific. During questions or a practice run, listeners need to refer back to a slide quickly (e.g., "On slide 14, how was that variable defined?") instead of describing it or guessing where a topic occurred.
- **Complete, record, and review practice runs**: Before any significant presentation, rehearse the complete talk out loud, record the practice run (with screen recording software, a webcam, or your phone), and watch the recording. The recording shows your pacing, timing, filler words, and transitions, and whether slide text and figures stay legible on screen.
- **One main point per slide**: Limit the cognitive load of each slide. If a slide presents several findings or a complex argument, split it across consecutive slides so the audience can follow each point.
- **Prefer visual evidence**: Audiences cannot read dense paragraphs while listening to a speaker. Use clean charts, diagrams, and annotated figures instead of text-heavy bullet lists.
- **Ensure high contrast and readability**: Use large fonts and high-contrast colors. Check that axis labels, legends, and annotations can be read from the back of a large room or on a low-resolution projector.

Presentation and poster resources

- How to tell a compelling story in scientific presentations (Kirchoff 2021)
- The science of storytelling: the David Attenborough style of scientific presentation (Ratcliff 2023)
- How to create a better research poster in less time - Mike Morrison's video introducing the #betterposter template
- Towards an "#evenbetterposter" - the MIT Biological Engineering Communication Lab's improvements to that template

Scientific figures

Include a figure wherever one helps a reader see a definition, an example, or a result. Draw a graph for a relationship between quantities, a diagram for a structure or a process, and a plot of the data for a fitted model or an estimate. Add figures to new work as you write it, and add them to existing chapters that lack them. On a website or slide deck, draw a graph as an interactive plotly figure where feasible, and as a 3D plotly figure where a third dimension shows the idea, such as a surface over two parameters. Keep a static image for the PDF and Word outputs, which cannot show a widget. Give every figure alt text, a numbered caption, and a code chunk that comments what the code does and why.

A clear figure lets a reader check a result without help from the text. Before you share a figure:

- Label the x-axis and y-axis clearly.
- Use panel labels when a figure has several subplots.
- Include enough tick marks and tick labels.
- Print tick labels as plain or comma-separated numbers, such as 0.001 or 10,000, not in scientific notation. A logarithmic axis is fine. Only the label format matters, so write 1, 10, 100 and 1,000, not 1e+00 through 1e+03. When the numbers are huge or tiny, rescale the units and say so in the axis title, such as "dollars (millions)".
- Use font sizes that are easy to read without dominating the plot.
- Use colorblind-friendly palettes (see ColorBrewer or iwanthue).
- Define colors, shapes, and line types in a legend.
- Minimize abbreviations in labels and legends.
- Use transparency to show overlapping data.
- Show measures of precision, such as 95% confidence intervals.

For more:

- Ten Simple Rules for Better Figures (Rougier et al. 2014)
- Tufte CSS - style HTML articles like Edward Tufte's books and handouts, with sidenotes, typography, and tight integration of graphics with text

Three-dimensional graphs

A function of two variables, such as a loss or a residual sum of squares as a function of a regression's intercept and slope, is a surface. A contour plot shows where the function takes given values and is easy to read values from. A three-dimensional surface plot shows the same function as a landscape, so valleys, ridges and minima are visible at once. Show both when the shape of the surface matters.

- Use a 3D plot for a surface, with the function's value on the vertical axis, and pair it with a contour plot of the same function.
- Do not use 3D for data that are not a surface, such as bars or pie charts: perspective makes lengths and areas harder to compare.
- Choose the viewing angle so that the feature you discuss, such as the minimum, is not hidden, and label all three axes.
- An interactive plot that readers can rotate, such as one made with plotly, helps on the web, but PDF and Word output show no interactive plot, so also provide a static view at a fixed angle (see Make figures accessible).
- Give a 3D plot alt text that states what the surface looks like, such as "a bowl with its lowest point at …".

> **Source**
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> The sections on slide decks, presentation resources and scientific figures are adapted from the Professional Development and Resources chapters of the UCD-SERG lab manual. Several of the resource links began in the Benjamin-Chung Lab Manual by Jade Benjamin-Chung. The section on three-dimensional graphs is original to this guide.

Kirchoff, Bruce. 2021. "How to Tell a Compelling Story in Scientific Presentations." *Nature* 600 (7890): S88–89. https://doi.org/10.1038/d41586-021-03603-2.

Ratcliff, William C. 2023. "The Science of Storytelling: The David Attenborough Style of Scientific Presentation." *Trends in Molecular Medicine* 29 (9): 675–77. https://doi.org/10.1016/j.molmed.2023.05.002.

Rougier, Nicolas P., Michael Droettboom, and Philip E. Bourne. 2014. "Ten Simple Rules for Better Figures." *PLOS Computational Biology* 10 (9): e1003833. https://doi.org/10.1371/journal.pcbi.1003833.